

Faculty	Engineering And Technology		
Assessment Name (Eg: End/Sup)	END ASSESSMENT	Paper Code	CADJ/A25-S
Module Name	Advanced Java	Module Code	C7-ADJ-11
Semester (E.g Jul-Dec-2021)	Jan-Jun 2025	Submission Mode (E.g Turnitin, blackboard, hardcopy)	turnitin
Total Marks	100	Duration (E.g, mins/hours/days/weeks)	1 week
Assessment Type (Eg: Written/Practical/Sub mission)	Submission	Submission Date (Applicable for submission-based exams only)	23rd June 2025 23:59

Instructions

1. Late submissions will not be accepted.
2. Export your database along with sample data inserted in your tables.
3. Your database name should be your Student Number.
4. Your project name should start with your student Number.
5. Add a readme file to your project explaining how to setup your project and all default user names and passwords since password ought to be stored in their hashed form.
6. If using any external jars, make sure to include such jars in your project folder.
7. The cover page of your report ensure to write all your particulars and batch code.
8. Submission of the project artefact is on Blackboard.
9. Submission of the project report is on turnitin.
10. Similarity index should not exceed 15%.
11. Ensure academic honesty, any kind of misconduct will result in disciplinary action as per the university's academic integrity policy, which may include a reduction in marks, failure of the project, or further penalties as deemed appropriate
 - Usage of AI tools in this assessment is not allowed.

Internship Management System – Project Specification

Overview

As a student who has just completed an internship, you have experienced firsthand the challenges of finding, applying for, and tracking internship opportunities. Many students struggle with unclear application processes, delays in employer responses, and the lack of a structured system to provide feedback on their experiences.

To address these challenges, you have partnered with your university to develop a web-based Internship Management System that will allow students to easily browse internships, apply, upload necessary documents, receive notifications, and provide reviews. The system will also help employers manage applications efficiently and enable the university to oversee the process to ensure fair opportunities for all students.

Your task is to develop this Internship Management System using Java Servlets and the MVC architecture, ensuring a structured, maintainable, and scalable platform.

Functional Requirements

As part of this project, you are required to implement the following core functionalities:

1. User Authentication & Role-Based Access

- a. The system should support three types of users: Students, Employers, and Administrators.
- b. Users must log in with their credentials, and based on their role, they should be redirected to their respective dashboards.
- c. Session management should be implemented to prevent unauthorized access.

2. Internship Listings with Pagination

- a. Students should be able to browse available internships with a paginated view (e.g., 5 internships per page).
- b. Each internship listing should display:
 - Title
 - Company Name
 - Location
 - Posted Date
 - Brief Description
 - Apply Now Button
- c. Employers should be able to post, update, and remove internships.

3. Internship Applications & Document Upload

- a. Students should be able to apply for internships by submitting an application form.
- b. During the application, students must upload their CV and/or academic transcript in PDF format.
- c. Employers should be able to view and download these documents when reviewing applications.

4. Email Notifications

- a. The system should send automated email notifications to students when:
 - i. They successfully apply for an internship.
 - ii. Their application status changes (Accepted/Rejected/Pending).
 - iii. A new internship relevant to their interests is posted.

5. Review & Feedback System (with Employer Replies)

- a. After completing an internship, students can submit a review with:
 - i. A rating (1-5 stars)
 - ii. Comments on their experience
 - iii. Submission date
- b. Employers should be able to reply to reviews, providing additional feedback or addressing concerns.
- c. Administrators should have the ability to moderate reviews and remove inappropriate content.

Website Properties & User Flow

Website Characteristics

- **User-Friendly Interface:** The website should have a clean and intuitive design, ensuring a smooth experience for students, employers, and admins.
- **Secure Authentication:** Role-based access should prevent unauthorized users from performing restricted actions.
- **Scalable Architecture:** Built using Java Servlets, JSP, and MySQL, following the MVC pattern for maintainability.
- **Session Management:** Users should stay logged in until they manually log out or their session expires.
- **File Management:** Uploaded CVs and Transcripts should be securely stored and accessible only to authorized users.

Role-Based Functionalities

1. Student Role

- Register/Login
- View & Browse Internship Listings (with Pagination)
- Apply for Internships and Upload CV/Transcript
- Track Application Status & Receive Email Updates
- Submit Internship Reviews & Ratings

2. Employer Role

- Login to the Employer Dashboard
- Post, Edit, and Remove Internship Listings
- Review & Manage Internship Applications
- Download Student CVs and Transcripts
- Accept/Reject Internship Applications (sending email notifications)
- Reply to Student Reviews

3. Administrator Role

- Login to the Admin Panel
- Manage User Accounts (Activate/Deactivate Users)
- Monitor & Moderate Reviews (Delete inappropriate content)
- Oversee System Activities & Ensure Smooth Operations

User Flow for Each Role

1. Student User Flow

- Login** → Enters the system and lands on the **student dashboard**.
- Browse Internships** → Uses pagination to view internship listings.
- Apply for Internship** → Clicks “Apply” → Fills out application form.
- Receive Email Notification** → Confirms application was submitted.
- Check Application Status** → Gets notified when application status changes.
- Submit Review** → After internship completion, writes a review.

2. Employer User Flow

- Login** → Lands on the **employer dashboard**.
- Post an Internship** → Fills out the internship details form.
- Manage Applications** → Views student applications and updates their status.
- Reply to Reviews** → Engages with student feedback by responding to reviews.

3. Admin User Flow

- Login** → Lands on the **admin panel**.
- Manage Users** → Can view, activate, or deactivate accounts.
- Moderate Reviews** → Can delete inappropriate reviews.
- Monitor System Activities** → Ensures smooth operation of the platform.

Section B : Report and Documentation [30]

This section assesses the students' ability to document their project, justify their design choices, and research the technologies used in their implementation, hence the report should include:

1. Introduction & Problem Statement [5 Marks]
2. System Design & Architecture [20 Marks]
 - A. Implementation of MVC Using Servlet Technology
 - Students should explain how they implemented MVC using **Servlets, JSP, and Java Beans**. (6)
 - With the help of a diagram explain servlet technology and its architecture. (5)
 - Explain how the MVC architecture ensures separation of concerns. (4)
 - B. Database Design & Entity Relationships
 - Create a well-structured ER diagram showing all entities within your DB in 3NF. (5)
3. Research & References [5 Marks]
 - (3 Marks) Citing academic papers, official documentation, or industry sources to support choices.
 - (2 Marks) Proper referencing in APA/Harvard format for all sources used

Marking Guide For Section A

1. User Authentication & Role-Based Access (10 Marks)

- **(3 Marks)** User registration & login functionality (Students, Employers, Administrators).
- **(3 Marks)** Session management (maintaining login state, preventing unauthorized access).
- **(2 Marks)** Role-based redirection (Each user is directed to their respective dashboard).
- **(2 Marks)** Secure password handling (hashed passwords, validation).

2. Internship Listings with Pagination (8 Marks)

- **(4 Marks)** Internships are displayed with necessary details (Title, Company Name, Location, Posted Date, Brief Description, Apply Button).
- **(4 Marks)** Pagination implemented correctly (e.g., 5 internships per page, smooth navigation).

3. Internship Applications & Document Upload (10 Marks)

- **(4 Marks)** Students can successfully submit internship applications.
- **(3 Marks)** CV/Transcript upload is functional, secure, and supports only PDF format.
- **(3 Marks)** Employers can **view and download** uploaded CVs and transcripts.

4. Email Notifications (7 Marks)

- **(2 Marks)** Students receive emails upon successful application submission.
- **(3 Marks)** Students receive email updates when application status changes (Accepted/Rejected/Pending).
- **(2 Marks)** Email notifications for new internships posted (optional, but bonus points if implemented).

5. Review & Feedback System (8 Marks)

- **(3 Marks)** Students can submit reviews with a rating (1-5 stars) and comments.
- **(3 Marks)** Employers can **reply to reviews** with feedback.
- **(2 Marks)** Administrators can moderate or delete inappropriate reviews.

6. Employer Functionalities (7 Marks)

- **(2 Marks)** Employers can create, edit, and remove internship listings.
- **(2 Marks)** Employers can review and manage student applications.
- **(3 Marks)** Employers can accept/reject applications with status updates.

7. Administrator Functionalities (6 Marks)

- **(2 Marks)** Admins can activate/deactivate user accounts.
- **(2 Marks)** Admins can oversee system activities (view applications, internships).
- **(2 Marks)** Admins can moderate reviews and delete inappropriate ones.

8. Website Structure & MVC Implementation (7 Marks)

- **(3 Marks)** Proper use of MVC architecture (Separation of concerns: Model, View, Controller).
- **(2 Marks)** Well-structured database integration.
- **(2 Marks)** Code readability, maintainability, and best practices.

9. General System Functionality & User Experience (7 Marks)

- **(3 Marks)** System is fully functional and meets all basic requirements.
- **(2 Marks)** User-friendly interface with intuitive navigation.
- **(2 Marks)** Proper error handling (invalid logins, missing fields, security vulnerabilities handled).

Good Luck